

kLCRNet: Fast Road Network Extraction via Keypoint-Driven Local Connectivity Exploration

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Abstract—Road network extraction from remote sensing images has been extensively studied in recent decades. While many approaches output road networks in vector format, most are not fully end-to-end, requiring time consuming postprocessing steps. In addition, challenges like isomorphic encoding limit the flexibility of these methods. In this article, we present kLCRNet, an efficient road network extraction framework that overcomes these limitations by leveraging keypoint-driven local connectivity exploration. kLCRNet consists of two key components: A keypoint detection module that identifies road keypoints via heatmap-based detection and refines them using bipartite matching, and a local connectivity exploration module that samples local connection relationships to directly construct connectivity between detected keypoints. Experiments on the CityScale and SpaceNet datasets demonstrate that kLCRNet outperforms state-of-the-art methods in topological accuracy and connectivity. In addition, kLCRNet significantly improves inference speed by up to 25 times, highlighting its efficiency and effectiveness.

Index Terms—Keypoint detection (KPD), local connectivity, remote sensing, road network extraction.

I. INTRODUCTION

ROAD networks are essential for a wide range of real-world applications, including autonomous driving, navigation, disaster response, and urban planning [1], [2]. Companies like Google Maps rely on street view vehicles equipped with cameras, radar, and GPS to gather high-precision road data [3]. While these methods provide accurate results, they are labor-intensive and time consuming, especially when covering large areas. In contrast, deep learning-based road extraction methods [4], [5], [6], [7], [8], [9] offer a more efficient alternative by automatically extracting road networks from high-resolution satellite imagery. These approaches are not only cost-effective but also provide high accuracy with timely updates. Consequently, research into deep learning models for road network extraction from remote sensing images holds great potential to enhance disaster relief operations and support numerous other applications.

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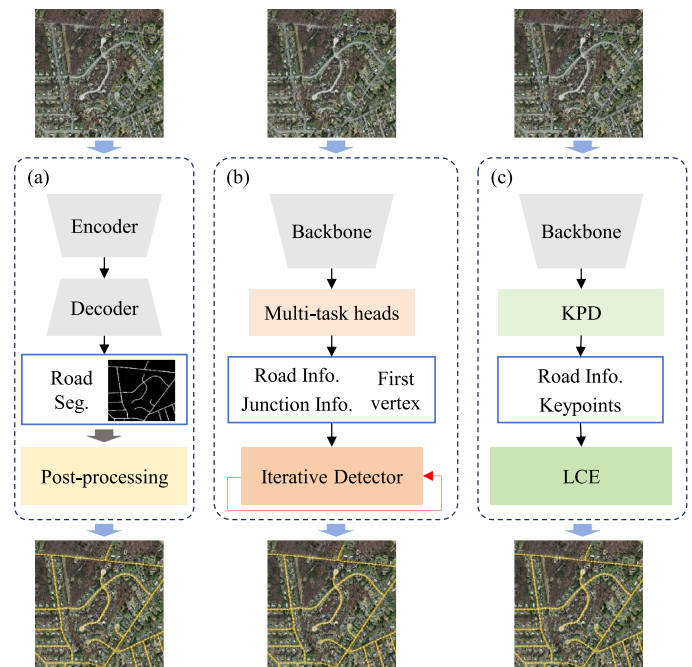


Fig. 1. Different methods for road network extraction include the following: (a) Semantic segmentation-based methods require post-processing for vectorization and refinement, which is not end-to-end and suboptimal. (b) Graph-based methods employ iterative detectors to extract road networks in a vector format, suffering from occlusions and time-consuming. (c) kLCRNet introduces a novel framework by combining a keypoint detection (KPD) module with a local connectivity exploration (LCE) module, enabling efficient and end-to-end road extraction.

Traditional deep learning methods [10], [11], [12], [13] for road extraction rely on semantic segmentation, which generates a binary segmentation map by classifying each pixel as either road or nonroad. While these approaches can automatically extract road masks from remote sensing images, they face significant challenges due to occlusions, shadows, and variations in image quality. Moreover, road networks in real-world applications are typically represented as graph structures with vertices and edges in a vector format [14]. As illustrated in Fig. 1(a), these methods often require post-processing steps to vectorize and refine road networks [15], [16]. Consequently, semantic segmentation-based road extraction is not well-suited for practical applications where topological accuracy and efficiency are critical.

Graph-based methods [17], [18], [19] are gaining increasing attention for road network extraction due to their ability to

preserve connectivity, as shown in Fig. 1(b). These approaches model road networks as graph structures and use this representation as a supervisory signal during model training. Hence, this line of works effectively avoids the connectivity loss inherent in road masks produced by semantic segmentation. In addition, these methods generate road networks directly in a vector format, eliminating the need for complex postprocessing steps and reducing potential information loss. By modeling road networks as graphs, they can accurately capture complex topological features, such as overpasses, offering a distinct advantage for urban road network extraction.

In the exploration of graph-based road extraction methods, several innovative research directions have emerged. One category involves iterative-based detection approaches [18], [20], [21], where an iterative detector gradually extends from an initial point to cover the entire road network. However, these step-by-step methods are time consuming, limiting their scalability and suitability for large-scale or real-time applications. Another direction focuses on graph vector encoding [22], [23], which encodes road networks as keypoints and corresponding offset vectors. These methods facilitate road network extraction through a noniterative encoding process. During inference, these vectors establish connectivity, offering improved efficiency. However, these methods still rely on time-consuming post-processing for decoding, and challenges like isomorphic encoding reduce the model's flexibility in accurately representing road networks. This work addresses these limitations by proposing a novel solution for more efficient and direct road network extraction.

In this study, we focus on a core characteristic of road network: connectivity within a local spatial range. We represent road networks as graph structures, consisting of keypoints and their interconnections. These keypoints and their local connectivity are crucial for maintaining the integrity and accuracy of road networks. Based on this perspective, this article introduces kLCRNet, an efficient road network extraction framework that leverages the local connectivity of keypoints. As shown in Fig. 1(c), kLCRNet emphasizes defining the interconnections between keypoints in road networks. Firstly, kLCRNet detects and matches keypoints in remote sensing images. A detection module generates road keypoints from a heatmap, followed by a nonmaximum suppression (NMS) step to refine results. Subsequently, a bipartite matching mechanism is applied to accurately align predicted keypoints with the ground truth. Second, kLCRNet introduces a local connectivity exploration module to establish connection relationships. This module simplifies the complex task of connection construction by converting it to a binary classification problem. It systematically enumerates all potential connections within the local spatial range of each keypoint, sampling ground truth connections as positive examples and the remainder as negative. Finally, a line pooling operation extracts features from these samples, which are classified through a linear layer to complete the road network extraction process.

Extensive experiments on the CityScale [22] and SpaceNet [24] datasets highlight the notable advantages of kLCRNet. The method exhibits proficiency in modeling complex road structures, such as overpasses, manifesting discernible advantages

in urban road network extraction scenarios. In summary, our contributions are concluded as follows.

- 1) We propose kLCRNet, an efficient framework for road network extraction from remote sensing images. kLCRNet models road networks as graph structures and establishes local connectivity between keypoints to construct road networks.
- 2) kLCRNet employs a two-stage architecture. In the initial stage, it utilizes a keypoint detection module to generate keypoints from a heatmap, followed by a bipartite matching operation for keypoint alignment. The subsequent stage designs a local connectivity exploration module that focuses on establishing connectivity between keypoints.
- 3) Experiments on two public dataset evaluate kLCRNet in terms of precision, topology, and inference speed. kLCRNet outperforms state-of-the-art methods and effectively handles complex urban scenes, demonstrating its potential as a valuable tool for road network extraction.

The rest of this article is organized as follows. Section II reviews related studies. Section III presents a detailed introduction to our proposed kLCRNet. Section IV describes the experiment settings, datasets, and evaluation metrics, and then reports and discusses the experimental results both quantitatively and qualitatively. Section V analyzes kLCRNet in terms of model structure, performance, limitations, and future work. Finally, Section VI concludes this article.

II. RELATED WORK

Road network extraction is critical in remote sensing image analysis. Early methods [25], [26] utilized deep learning to automatically extract roads from extensive remote sensing data, significantly enhancing the efficiency and accuracy of road network extraction. This advancement has driven ongoing evolution in the field. In this section, we mainly discuss two prominent categories of road extraction methods that are relevant to our work.

A. Semantic Segmentation-Based Road Extraction

Semantic segmentation-based road extraction methods leverage advanced semantic segmentation techniques [27], [28], [29] to extract road networks from remote sensing images by classifying each pixel as either road or nonroad. Zhang et al. [5] introduced a Res-UNet network, which combines an encoder-decoder structure with the residual connections of ResNet [30]. The Res-UNet network enhances convolutional modules in both the encoder and decoder stages. This improvement significantly boosts the accuracy of road semantic segmentation. Zhou et al. [31] developed a D-LinkNet model, integrating LinkNet [32] with dilated convolution operations to expand the model's receptive field, resulting in superior road network segmentation while optimizing computational efficiency. Due to their simplicity and high recall characteristics, these methods achieve higher accuracy in road extraction compared to traditional techniques. However, they primarily generate road masks and focus on pixel-wise segmentation accuracy. As a result, they

often overlook the connectivity within road networks and may not fully meet the requirements of practical applications.

Another line of semantic segmentation-based methods represents the task as a multistage process. Initially, these approaches achieve high recall in road area segmentation through pixel-level classification. They then apply postprocessing techniques, such as morphological operations and skeleton extraction, to generate vectorized road networks. To address issues of disconnection, Mátyus et al. [17] used semantic segmentation for initial results and employed a shortest path algorithm to infer missing connections in the road network. Inspired by manually road annotation in a specific orientation, Batra et al. [33] propose a connectivity task that jointly learns orientation and segmentation, enhancing the accuracy of road topological maps. To tackle fragmented roads, Mei et al. [19] developed a connectivity attention network. This network uses strip convolutions to capture long-range context from various directions while minimizing interference from irrelevant background. They also introduce a connectivity attention module to explore relationships between neighboring pixels and use graphical information to preserve road connectivity. Zhou et al. [34] proposed a split depth-wise separable graph convolutional network to separately capture channel and spatial features, improving road representation learning. Dai et al. [35] introduced a road-augmented deformable attention network that captures long-range dependencies through strip convolutions and a deformable attention module, enhancing feature extraction by leveraging spatial and shape correlations. While these multistage methods improve road connectivity and segmentation accuracy, they often involve complex computations and may not be well suited for processing large-scale remote sensing images.

B. Graph Modeling-Based Road Extraction

Graph modeling-based road extraction has gained significant attention for its ability to explicitly construct connectivity information of road networks. These approaches [36], [37], [38] represent road networks as a graphs, where vertices denote keypoints and edges represent the connections between these keypoints. By directly generating road networks in vector format, graph modeling-based approaches avoid the connectivity loss associated with segmentation and the information loss typically encountered in postprocessing. In addition, these methods can model complex road networks, making them especially beneficial for extracting road networks in urban scene.

The predominant approach in graph modeling-based road extraction is iterative detection. These methods detect the road network step by step. Bastani et al. [18] pioneered this approach by employing a convolutional neural network to detect road extensions from a given keypoint, enabling gradual inference of the entire road network. Tan et al. [20] proposed a point-based iterative detection method that leverages segmentation guidance and initializes the detection process with segmentation results, effectively enhancing road network recall and connectivity. Xu et al. [21] developed an end-to-end iterative detector based on DETR [39], significantly improving road network accuracy. Furthermore, Xu et al. [40] integrated road instance segmentation and multiscale feature fusion to achieve even greater accuracy.

While these iterative detection methods preserve connectivity and enhance accuracy, they are time-consuming and limited for large-scale applications. In addition, the iterative nature of these methods can lead to error accumulation throughout the pipeline, resulting in a higher number of false positive predictions.

To advance road network extraction methods, several novel approaches have been proposed. He et al. [22] introduced a graph-tensor encoding method that represents the road network using keypoints and offset vectors. This approach constructs road network connectivity with offset vectors during inference. This approach significantly improves both extraction speed and connectivity accuracy. Zao et al. [23] propose TopoRoad, which decomposes road network extraction into multiple subtasks, including vertex detection, vertex degree estimation, direction prediction, and segmentation. These tasks are then integrated into a road network using a decoding module, achieving precise extraction from remote sensing images. SAMRoad [41] first adapts the Segment Anything Model (SAM) [42] to the road extraction task. It fine-tunes the image encoder to segment roads and intersections, then uses a transformer-based decoder to generate edges. However, these methods are not fully end-to-end and still rely on time consuming postprocessing. Therefore, optimizing model structures to minimize postprocessing time and enhance end-to-end performance remains a crucial area for future research.

III. METHOD

In this section, we present kLCRNet, an efficient road network extraction framework that leverages local connectivity exploration of keypoints. We will describe the overall pipeline and key components of kLCRNet in detail.

A. Overview

Graph modeling-based road extraction methods have recently advanced through iterative detection and graph-tensor encoding, effectively preserving road segments and connection relationships. While these approaches enhance connectivity and accuracy, they are often time consuming and lack end-to-end processing due to iterative steps or postprocessing requirements. To address these challenges, kLCRNet offers a solution by extracting road networks through keypoint-driven local connectivity exploration in a fully end-to-end manner.

As shown in Fig. 2, kLCRNet utilizes a two-stage architecture to directly generate the road network graph G from an input image I . In the first stage, a CNN-based backbone extracts features from the input, and keypoints are detected from the feature map F . In the second stage, the connectivity between keypoints is established. Specially, the road network graph is denoted as $G = (V, E)$, where V represents the set of predicted keypoints from the first stage, and E denotes the edge set representing road connections from the second stage. This approach ensures a seamless, integrated process for road network extraction, enhancing both efficiency and accuracy.

In stage 1, kLCRNet incorporates two branches: 1) a keypoint detection branch; and 2) a road segmentation branch. These branches work concurrently to perform keypoint detection and

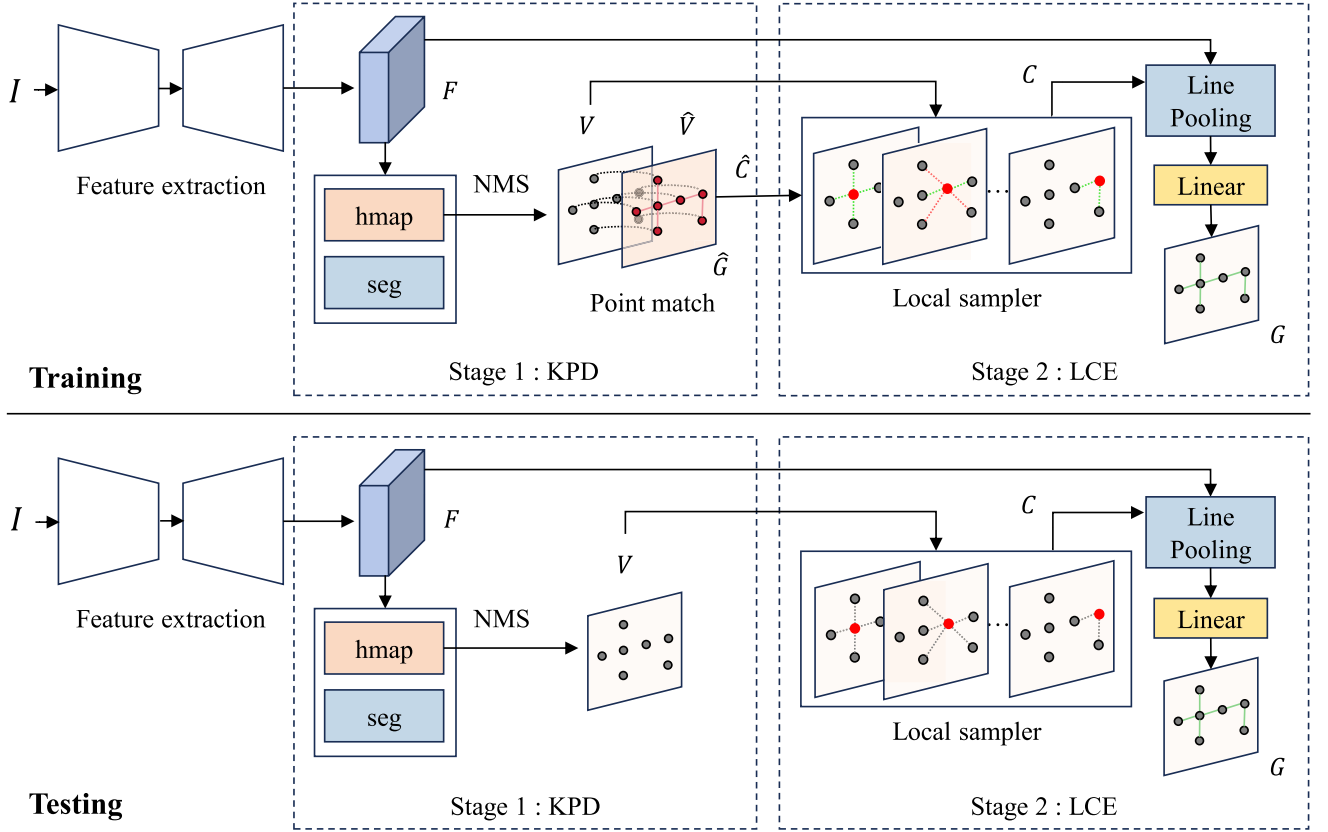


Fig. 2. Framework of kLCRNet, an efficient road network extraction based on local connectivity exploration of keypoints. kLCRNet is a two-stage architecture comprising a keypoint detection (KPD) module and a local connectivity exploration (LCE) module. The KPD module is responsible for detecting keypoints, while the LCE module focuses on exploring and establishing local connectivity between these keypoints.

road segment segmentation. An NMS operation is applied to generate the keypoint prediction set V . During training, keypoint matching aligns the prediction set V with the ground truth $\hat{V}$, resulting in a refined keypoint prediction set V .

In stage 2, kLCRNet uses the final keypoint prediction set V and the ground truth connectivity $\hat{C}$ as inputs. The proposed local connectivity exploration module then constructs positive and negative samples (C_{pos}, C_{neg}) of the connectivity C between keypoints. Subsequently, a line pooling operation is applied to align feature lengths from the feature map F . These aligned features are then classified into connectivity samples using a linear layer. This process ensures accurate construction and classification of connectivity within road networks.

B. Feature Extraction

In the feature extraction module, we use a CNN-based backbone for its lightweight and easy of deployment, while also meeting the model's performance and time requirements. Unlike general keypoint detection, road keypoints are denser, necessitating a backbone with high resolution and rich semantic information. Thus, we employ HRNet [43] to extract feature map F . HRNet [43] is advantageous because it fuses multiscale features through parallel branches with varying resolutions, enhancing both semantic and spatial information in the feature maps. The downsampling factor of the output feature map F is denoted as

r , which is further discussed in Section IV-E. The ground truth $\hat{G}$ is downsampled by a factor of r to provide the supervision signal, while the predicted graph G is upsampled by the same factor r to generate the final output.

C. Keypoint Detection

In stage 1, the keypoint detection module first designs multitask detection branches to generate road keypoints and masks. Redundant keypoints are then filtered out using an NMS operation. Finally, the Hungarian algorithm [44] is applied for road keypoint matching.

Multitask detection branches: The keypoint detection module leverages multitask learning to simultaneously extract keypoint heatmaps and road masks. The keypoint heatmaps are used to generate the final road keypoints and road masks with rich semantic information serves as crucial input for stage 2.

Given the input remote sensing image $I \in \mathbb{R}^{H \times W \times 3}$, the keypoint heatmap head predicts a heatmap $Y \in [0, 1]^{H_r \times W_r}$, where H , W , and r denote the height, width, and downsampling coefficient, respectively. The heatmap Y represents the likelihood of each position being a road keypoint. Road keypoints are defined as vertices interpolated at uniform intervals along a road. During training, the ground truth heatmap $\hat{Y}$ is generated

using a Gaussian kernel function as follows:

$$\hat{Y}_{yx} = \exp\left(-\frac{(x - \tilde{x})^2 + (y - \tilde{y})^2}{2\sigma^2}\right) \quad (1)$$

where $\tilde{x}$ and $\tilde{y}$ are the coordinates of each keypoint, and the standard deviation σ is related to the spatial resolution of the input remote sensing image. Specially, when the heatmaps generated by two overlapping keypoints intersect, the maximum value is preserved at the overlapping regions.

Given the sparsity of road keypoints in remote sensing images, we use Focal Loss [45] to address the imbalance between keypoint areas and nonkeypoint areas, which is denoted as follows:

$$L_{kpt} = \frac{-1}{H' \times W'} \times \sum_{yx} \begin{cases} (1 - Y_{yx})^\alpha \log(Y_{yx}) & \hat{Y}_{yx} = 1 \\ (1 - \hat{Y}_{yx})^\beta (Y_{yx})^\alpha \log(1 - Y_{yx}) & Y_{yx} \neq 1 \end{cases} \quad (2)$$

where α and β are the hyperparameters of Focal Loss, $H' \times W'$ denotes $H/r \times W/r$, and the subscript yx represents the coordinate (x, y) . Due to the downsampling factor r , there may be discretization errors in the predicted positions of keypoints on the heatmap. To improve the accuracy of these positions, the heatmap head also predicts a keypoint offset map $O \in \mathbb{R}^{H/r \times W/r \times 2}$, which measures the offset of each keypoint relative to its center. Finally, we apply L_2 loss to the keypoint offset map O prediction, defined as follows:

$$L_{off} = \frac{1}{\hat{N}} \sum_{\tilde{v}} \left\| O_{\tilde{v}} - \left(\frac{\tilde{v}}{r} - \left\lfloor \frac{\tilde{v}}{r} \right\rfloor \right) \right\|_2 \quad (3)$$

where $\hat{N}$ represents the total number of keypoints, and $O_{\tilde{v}}$ denotes the predicted offset for the keypoint $\tilde{v}$. In the keypoint offset loss function, only the positions $\tilde{v}$ where keypoints are present are considered for loss calculation, while positions without keypoints are ignored. In addition, the road segmentation head uses cross-entropy loss to predict road masks

$$L_{seg} = -\hat{P} \log(P) - (1 - \hat{P}) \log(1 - P). \quad (4)$$

Road keypoint filtering: To eliminate redundant keypoints, the keypoint detection module applies NMS to filter out nonmaximum values in the predicted heatmap. In practice, we use a 3×3 MaxPooling operation to perform this filtering in an end-to-end manner.

Since the prediction probabilities for keypoints are low during the initial stage of training, we employ a TopK algorithm to ensure a consistent and sufficient number of prediction points for subsequent keypoint matching, as described in Formula (5). During inference, a threshold KPT_THR is used to filter keypoints, as detailed in Formula (6)

$$V = \{(y + \delta_y, x + \delta_x) | Y'_{yx} \in TopK(Y'), (\delta_y, \delta_x) = O_{yx}\} \quad (5)$$

$$V = \{(y + \delta_y, x + \delta_x) | Y'_{yx} > KPT_THR, (\delta_y, \delta_x) = O_{yx}\}. \quad (6)$$

Road keypoint matching: To achieve optimal matching between predicted keypoints V and ground truths $\hat{V}$, we use the Hungarian algorithm [44]. The matching process assigns each ground truth keypoint to its best-matching prediction. The remaining unmatched predictions are treated as false positives and contribute to the loss function accordingly. To avoid additional penalties in the loss function due to incomplete matches, the keypoint prediction head generates a larger number of predicted keypoints, ensuring a strict one-to-one correspondence between predicted and ground truth keypoints. Moreover, a sufficient number of predictions can cover all possible ground truth keypoints, making it more robust in complex scenarios. This algorithm provides a bipartite matching $\pi \in \prod_{\hat{N}}^N$ from V to $\hat{V}$, where π represents permutations of N predicted keypoints with respect to $\hat{N}$ ground truths. Here, N denotes the number of predicted keypoints and $\hat{N}$ denotes the number of ground truths. The matching loss function is based on the Euclidean distance between matched keypoints

$$\text{cost}(v_i, v_j) = \|v_i - v_j\|. \quad (7)$$

So, the road keypoint matching problem can be formulated as

$$\pi = \arg \min_{\pi \in \prod_{\hat{N}}^N} \sum_i^N \text{cost}(\hat{v}_i, v_{\pi(i)}), \hat{v}_i \in \hat{V}, v_{\pi(i)} \in V \quad (8)$$

where $\pi(i)$ represents the i th value of the permutation π .

Since the keypoint detection stage can be unstable during training, optimal matching of road keypoints may suffer from high loss values. To address this issue, we set a matching loss threshold $MATCH_THR$ to include only those matches with losses below this threshold. In addition, we establish a maximum matching loss threshold MAX_MATCH_THR and a minimum matching threshold MIN_MATCH_THR . The threshold $MATCH_THR$ is designed to gradually decrease with training iterations to ensure more precise keypoint matching, as detailed below

$$MATCH_THR = \max(MIN_MATCH_THR, MAX_MATCH_THR \cdot 0.9^{ITER/10000}) \quad (9)$$

where $ITER$ represents the current training iteration.

D. Local Connectivity Exploration

The local connectivity exploration module is responsible for establishing connection relationships among keypoints V . This module ensures the integrity of the road network by focusing on both keypoints and their connectivity. It consists of two main components: 1) local connectivity sampling (LCS); and 2) line pooling.

Local connectivity sampling: As shown in Stage 2 of Fig. 2, the LCS constructs a connectivity set C by enumerating all possible connections between each keypoint within a specified local sampling radius

$$C = \{(v_i, v_j), l\} | v_i, v_j \in V, l \in \{0, 1\}\} \quad (10)$$

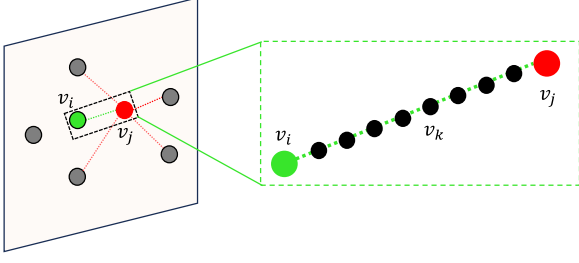


Fig. 3. Illustration of the line pooling operation for aligning features according to the connection relationship.

where v_i, v_j represents a pair of keypoints, l indicates the connectivity status with $l = 1$ for connected and $l = 0$ for disconnected, and $LOCAL_R$ denotes the local sampling radius. During training, LCS identifies the connectivity set $\hat{C}$ from the ground truth road network graph $\hat{G}$. Connections that match $\hat{C}$ are designated as positive samples C_{pos} (represented by green connections in Fig. 2, while the remaining connections are considered negative samples C_{neg} (represented by red connections in Fig. 2

$$\begin{aligned} C &= \{C_{pos}, C_{neg}\} \\ C_{pos} &= \{\{(v_i, v_j), l = 1\} | v_i, v_j \in V\} \\ C_{neg} &= \{\{(v_i, v_j), l = 0\} | v_i, v_j \in V\}. \end{aligned} \quad (11)$$

The LCS converts the connectivity exploration into a binary classification problem. During training, the local sampling radius $LOCAL_R$ must satisfy $LOCAL_R > MIN_LOCAL_R/r$, where MIN_LOCAL_R is the minimum distance between keypoints and r is the down-sampling coefficient. As shown in Fig. 2, during inference, the keypoint matching operation is skipped, and only the connectivity set is constructed.

Line pooling: The LCE module employs Line Pooling [46] to align features based on connection relationships. Given the feature map F , K points are uniformly sampled along each connection relationship $\{(v_i, v_j), l\} \in C$, as illustrated in Fig. 3. The sampling points are expressed as

$$v_k = \frac{k}{K-1}v_i + \frac{K-k}{K-1}v_j, k \in \{0, 1, \dots, K-1\}. \quad (12)$$

In mitigate errors caused by the quantization of floating-point coordinates, bilinear interpolation is applied to the feature map to obtain features at v_k . An average pooling operation (AvgPooling) is then performed to merge features of K sampled points. The aggregated feature is subsequently passed through a linear layer for classification. The connectivity classification loss is computed as follows:

$$L_{cls} = -\hat{l} \cdot \log(l) - (1 - \hat{l}) \cdot \log(1 - l). \quad (13)$$

In this article, kLCRNet utilizes multitask learning for road network extraction. The overall loss function integrates several components: Keypoint heatmap loss, keypoint offset map loss, road segmentation loss, and connectivity classification loss

$$L = \lambda_1 L_{kpt} + \lambda_2 L_{off} + \lambda_3 L_{seg} + \lambda_4 L_{cls} \quad (14)$$

where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are factors that adjust the relative influence level of different subtasks, which will be discussed in Section IV-E.

IV. EXPERIMENTS

This section describes the experimental datasets, evaluation metrics, implementation settings, and result analysis.

A. Dataset

We evaluate the model's performance on the CityScale dataset [22] and the SpaceNet dataset [24], comparing it with state-of-the-art methods. In addition, for the ablation study, we conduct experiments specifically on the CityScale dataset [22].

- 1) CityScale dataset [22] comprises 180 remote sensing images with a spatial resolution of 1 m, covering a total area of 720 km² across 20 cities in the United States. Each image has a resolution of 2048 × 2048. Road networks in this dataset are collected from OpenStreetMap (OSM) and serve as ground truth annotations. The dataset is partitioned into training, validation, and testing sets in a ratio of 144:9:27, following the split described in [22]. As shown in Fig. 4(a), OSM road networks are generated through user-contributed GPS tracks, providing a more authentic and reliable representation of real-world road conditions compared to manual annotations. Consequently, the CityScale dataset contains complex road structures, making it well suited for evaluating the performance of kLCRNet in extracting intricate road networks.
- 2) SpaceNet dataset [24] consists of 2549 remote sensing images, each with a spatial resolution of 30 cm and dimensions of 1300 × 1300. Road networks in the dataset are collected from OpenStreetMap (OSM). The dataset is split into training, validation, and testing sets with sizes of 2040, 127, and 382 images, respectively, as outlined in [22]. For consistency with the CityScale dataset, each image is downsampled to 400 × 400, resulting in an approximate spatial resolution of 1 m. The SpaceNet dataset includes imagery from four cities: Las Vegas, Paris, Shanghai, and Khartoum. As illustrated in Fig. 4(b), this dataset provides a relatively balanced representation of both urban and rural regions.

B. Evaluation Metrics

This article emphasizes the importance of connection relationship within road networks. While pixel segmentation accuracy is significant, the connectivity information is crucial for real-world applications. Therefore, we use TOPO [47] and APLS [24] as key evaluation indicators to assess the effectiveness of road network extraction methods.

TOPO: This indicator evaluates the similarity between graphs by analyzing their subgraphs. A subgraph is defined around a seed vertex, including vertices whose distance from the seed vertex is within a specified threshold. The similarity between subgraphs is determined by matching vertices, where two vertices are considered a successful match if they fall within a



Fig. 4. Examples from the evaluation datasets. Row (a): The CityScale dataset containing complex road structures, including overpasses. Row (b): The SpaceNet dataset covering a relatively balanced distribution of urban and rural regions.

predefined radius. TOPO operates by first randomly sampling seed vertices from both the ground truth and the predicted graphs. It then calculates the similarity of subgraphs derived from these seed vertices. For each seed vertex, TOPO computes the precision (TOPO-P), recall (TOPO-R), and F1 score (TOPO-F1) to quantify the similarity of the subgraphs. The calculation formulas for TOPO-P, TOPO-R, and TOPO-F1 are as follows:

$$\begin{aligned} \text{TOPO-P} &= \frac{V_M}{V_P}, \text{TOPO-R} = \frac{V_M}{V_G} \\ \text{TOPO-F1} &= 2 \times \frac{\text{TOPO-P} \times \text{TOPO-R}}{\text{TOPO-P} + \text{TOPO-R}}. \end{aligned} \quad (15)$$

TOPO-P, TOPO-R and TOPO-F1 differ from conventional evaluation metrics in how they assess graph similarity. Traditional methods focus primarily on comparing edge distances, often allowing vertices and edges of the graphs to be freely rotated to achieve the closest match. These conventional metrics typically emphasize topological similarity alone, without adequately considering the spatial configuration of the road networks. In contrast, TOPO incorporates both spatial and topological aspects when evaluating road networks. By analyzing subgraphs around seed vertices and ensuring that vertices match within a predefined radius, TOPO accounts for the actual spatial positioning of roads. This makes it a more comprehensive measure, as it reflects both the geometric layout and the network structure, offering a more realistic assessment for practical applications.

APLS: The APLS (Average Path Length Similarity) metric focuses on both the logical topology (the connections between vertices) and the physical topology (the actual layout of roads) within a road network. Unlike traditional pixel-based evaluation metrics, which primarily assess how closely predicted and ground truth pixels align, APLS emphasizes the connectivity and spatial arrangement of the network. APLS uses a Dijkstra's

shortest path algorithm to evaluate the distance between ground truth and predicted road networks

$$\text{APLS} = 1 - \frac{1}{n} \sum_{a,b \in V} \min \left\{ 1, \frac{|L(a,b) - L(\hat{a},\hat{b})|}{L(a,b)} \right\} \quad (16)$$

where a and b represent two randomly sampled vertices from ground truth, $\hat{a}$ and $\hat{b}$ denotes vertices of a and b mapped to the predicted graph. V indicates a vertex set of a road network and $L(\cdot)$ represents the shortest path calculation function. The APLS score ranges from $[0, 1]$, indicating that the extracted road network is from completely dissimilar to similar to ground truth.

Since the predicted road network may not be perfectly aligned with a real road network, APLS uses the vertex projection error radius to project vertices of ground truths onto the predicted network, ensuring that vertices are mostly aligned. In order to punish incorrect road intersection predictions, APLS not only projects vertices of ground truths onto a predicted network, but also performs an inverse operation (that is, projects vertices of a predicted road network onto ground truths). The shortcuts and breaks generated in the road network graph will affect the shortest path algorithm, causing APLS to produce low values. When the number of sampled vertices is large enough, APLS can reflect the connection relationship of a predicted graph. The APLS metric is nondifferentiable, which means it cannot be directly used in gradient-based optimization algorithms. The APLS metric is designed to better evaluate the performance of road network extraction, especially the accuracy of road network hub locations and intersections, which are critical for route planning and navigation.

TABLE I
EXPERIMENTAL RESULTS ON THE CITYSCALE DATASET

Method	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
UNet [28]	75.34	65.99	70.36	52.50
DeepRoadMapper [17]	76.54	71.25	73.80	54.32
Road-Orien [33]	75.83	68.90	72.20	55.34
RoadTracer [18]	78.00	57.44	66.16	57.29
Sat2Graph [22]	80.40	72.45	75.92	64.10
RNGDet [21]	85.97	69.78	76.87	65.57
TERNformer [48]	82.46	72.20	76.70	66.06
SAM-Road [41]	90.47	67.69	77.23	68.37
kLCRNet	86.77	71.59	78.27	68.96

C. Implementation Details

In practice, the HRNet-W32 backbone is employed for feature extraction, initialized with pre-trained weights. The model uses a down-sampling coefficient of $r = 2$. For the keypoint heatmap loss function, the parameters are set to $\alpha = 2$ and $\beta = 4$. In the keypoint selection process, the K value for the TopK algorithm is chosen as the sum of keypoint number plus 30, and the keypoint threshold KPT_THR is set to 0.25. Since the bipartite matching process benefits from having a higher number of predicted keypoints to ensure better coverage of all possible ground truth keypoints, particularly in complex scenarios, we have set the IoU threshold to 0.4 and the maximum number of retained keypoints to 200. These settings were chosen to strike a balance between keypoint retention and suppression, ensuring robust performance across various image complexities. During keypoint matching, the thresholds are set as $MIN_MATCH_THR = 1.1$ and $MAX_MATCH_THR = 3.0$. In the local connectivity exploration module, the sampling radius $LOCAL_R$ is set to 12.0, with a road network interpolation distance $MIN_LOCAL_R = 20$. For the line pooling operation, the number of sampling points K is set to 8, and the classification threshold CLS_THR is set to 0.4. The overall loss function is weighted with coefficients $\lambda_1 = 1.0$, $\lambda_2 = 1.0$, $\lambda_3 = 1.0$, and $\lambda_4 = 1.0$. The model is trained using the SGD optimizer with an initial learning rate of 0.02. A linear warm-up is applied for the first 4000 iterations, followed by a staged learning rate decay with a factor of 10 at 60 000 and 80 000 iterations. The batch size is set to 16, and the model is trained for 90 000 iterations in total.

D. Results and Discussion

Results on CityScale dataset: Experimental results on the CityScale dataset are reported in Table I. kLCRNet achieves the highest performance, with a TOPO-F1 score of 78.27% and an APLS value of 68.96%, demonstrating its superiority. Specifically, compared to the SOTA segmentation method TERNformer [48], kLCRNet improves the TOPO-F1 score by 1.57% and the APLS index by 2.90%. In addition, when compared with the iterative detection method RNGDet [21], kLCRNet outperforms by increasing the TOPO-F1 score by 1.40% and the APLS index by 3.39%. These results highlight the method's strength in capturing the topological accuracy of road networks.

The visualization results on the CityScale dataset are shown in Fig. 5. kLCRNet demonstrates strong performance across various scenarios, effectively capturing road network structures.

TABLE II
EXPERIMENTAL RESULTS ON THE SPACENET DATASET

Method	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
UNet [28]	68.96	66.32	67.61	53.77
DeepRoadMapper [17]	82.79	72.56	77.34	62.26
Road-Orien [33]	81.56	71.38	76.13	58.82
RoadTracer [18]	78.61	62.45	69.90	56.03
Sat2Graph [22]	85.93	76.55	80.97	64.43
RNGDet [21]	85.59	70.56	77.35	65.81
TopoRoad [23]	87.88	81.21	83.39	66.86
kLCRNet	89.69	72.20	80.00	67.51

TABLE III
QUANTITATIVE RESULTS ON THE MASSACHUSETTS ROADS DATASET

Method	kLCRNet	RNGDet [21]	Sat2Graph [22]	TERNformer [48]
TOPO-P $\uparrow$	94.13	67.57	85.88	90.75

The keypoint detection module significantly enhances the accuracy of road network extraction, resulting in a noticeable increase in TOPO-P, as it reduces false positives compared to the iterative detection method RNGDet [21]. Furthermore, kLCRNet achieves a substantially higher recall rate compared to graph modeling-based methods, indicating its effectiveness in accurately identifying road networks while minimizing missed connections.

Results on SpaceNet dataset: Table II presents the comparative results of different methods on the SpaceNet dataset. The proposed kLCRNet basically maintains a leading advantage in simpler road network scenarios, achieving a TOPO-P score of 89.69% and an APLS score of 67.51%. Although the TOPO-F1 is slightly affected by a lower recall, kLCRNet still secures the second-best result in this category. Despite a lower TOPO-R, the overall performance remains competitive, as reflected in Table II. kLCRNet exhibits clear advantages in terms of both logical and physical topology, surpassing other graph-based methods with the highest APLS value of 67.51%. The visualization results in Fig. 6 further highlight that kLCRNet produces more comprehensive road networks compared to alternative approaches, underscoring its efficacy in road network extraction.

Generalization on Massachusetts Roads dataset: To evaluate the generalization ability of kLCRNet, we conduct experiments on the Massachusetts Roads dataset [49], which covers urban, suburban, and rural areas offering a diverse range of data distribution scenarios. In this experiment, kLCRNet is trained on the CityScale dataset and tested on the Massachusetts Roads dataset. We compare the proposed method with existing approaches, including the iterative detection-based method RNGDet [21], the graph-based encoding method Sat2Graph [22], and the semantic segmentation-based method TERNformer [48]. Table III reports the experimental results, which demonstrate that kLCRNet achieves a higher TOPO-P of 94.13% across various data distributions. Fig. 7 shows visualization results, further indicating that kLCRNet is effective at modeling complex road structures. However, kLCRNet shows slightly inferior performance in some rural areas, which can be attributed to the fact that the ground truth in this dataset consists solely of pixel-level masks resulting in the loss of road network connectivity.



Fig. 5. Visualization results on the CityScale dataset. The cyan and orange lines represent the ground truth edges and the extracted edges, respectively. The yellow points indicate the extracted vertices. kLCRNet extracts road networks with good connectivity in most scenarios. Zoom in to see more detail. (a) Input. (b) RNGDet. (c) TERNformer. (d) kLCRNet. (e) Ground Truth.

TABLE IV
BASELINE RESULTS COMPARISON ON THE CITYSCALE DATASET

Setting	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
UNet [28]	75.34	65.99	70.36	52.50
kLCRNet	86.77	71.59	78.27	68.96
-seg	85.45	70.21	76.83	67.16
-decay of MATCH_THR	86.40	69.37	76.66	66.01

E. Ablation Study

This section presents ablation experiments to assess the impact of various parameter settings and verify the effectiveness of the proposed kLCRNet. The ablation studies are conducted on the CityScale dataset [22], which has complex road connection scenarios, making it an ideal benchmark to evaluate the robustness of different configurations in kLCRNet.

Baseline comparison: The baseline model employs the backbone for feature extraction followed by semantic segmentation, with UNet [28] serving as the baseline method. As shown in

Table IV, kLCRNet significantly improves both graph and connectivity metrics for road networks compared to this baseline. In the subsequent ablation study, we gradually remove the semantic segmentation head from the keypoint detection module and the *MATCH_THR* decay strategy from the keypoint matching process. The results demonstrate that both the inclusion of semantic segmentation information and the keypoint matching strategy play crucial roles in enhancing model performance.

The semantic segmentation head produces segmentation maps of road networks, which benefits the model in two key ways: 1) it enhances keypoint detection through multitask optimization; and 2) provides valuable semantic guidance to the classification network by concatenating with the feature map F . Incorporating the semantic segmentation head has led to notable improvements in performance, with the TOPO-F1 score increasing from 76.83% to 78.27% and the APLS value rising from 67.16% to 68.96%.

In the keypoint matching stage, kLCRNet employs the Hungarian algorithm for binary matching between predicted



Fig. 6. Visualization results on the SpaceNet dataset. The cyan and orange lines represent ground truth edges and the extracted edges, respectively. The yellow points represent the predicted vertices. kLCRNet generates more comprehensive road networks compared to other methods. Zoom in to see more detail. (a) Input. (b) RNGDet. (c) kLCRNet. (d) Ground Truth.

keypoints and ground truths. Given the large matching losses often encountered in early training stages, a fixed threshold for filtering matches can lead to suboptimal training of the connection relationship construction in stage 2. To address this issue, kLCRNet implements a *MATCH_THR* decay strategy, where the threshold gradually decreases over training iterations to improve keypoint matching accuracy. This approach has led to improvements in performance, with the TOPO-F1 score increasing from 76.66% to 76.83% and the APLS value increasing from 66.01% to 67.16%.

Backbone: Several CNN-based backbone networks have been evaluated in this experiment. ResUNet is a classic pipeline for semantic segmentation that combines ResNet [30] with a UNet [28] architecture. It leverages the residual connections of ResNet within the UNet framework to enhance segmentation performance. DLA-dcn [50] is an improved version of the DLA [52] network, which incorporates deformable convolutional layers to enhance keypoint detection capabilities. In addition, Higher-HRNet [51] is a variant of HRNet [43] designed for keypoint detection. It uses transposed convolution to



Fig. 7. Visualization on the Massachusetts Roads dataset. kLCRNet is trained on the CityScale dataset and tested on the Massachusetts Roads dataset. Zoom in to see more detail. (a) Input. (b) Sat2Graph. (c) TERNformer. (d) kLCRNet. (e) Ground Truth.

TABLE V

ABLATION STUDY OF DIFFERENT BACKBONES ON THE CITYSCALE DATASET

Backbone	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
ResUNet [28]	86.97	65.22	74.19	65.36
DLA-dcn [50]	87.49	70.24	77.66	65.99
Higher-HRNet [51]	86.77	71.59	78.27	68.96

TABLE VI

COMPARISON OF DOWN-SAMPLING FACTOR r ON THE CITYSCALE DATASET

r	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
1	88.56	68.90	77.21	65.82
2	86.77	71.59	78.27	68.96
3	86.33	65.82	74.35	62.19

TABLE VII

COMPARISON OF SAMPLING RADIUS $LOCAL_R$ ON THE CITYSCALE DATASET

$LOCAL_R$	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
10	83.14	72.44	77.24	69.56
12	86.77	71.59	78.27	68.96
14	88.13	70.58	78.18	68.29
16	88.57	69.80	77.81	66.99

maintain higher resolution feature maps, improving the precision of keypoint localization.

Experimental results are presented in Table V. The findings demonstrate that Higher-HRNet [51] outperforms other backbones, achieving a TOPO-F1 score of 78.27% and an APLS score of 68.96%. Unlike typical keypoint detection tasks, road keypoints benefit from higher resolution to preserve connectivity, which can be compromised by downsampling operations.

Downsampling factor r : Feature map resolution plays a crucial role in road network extraction. To identify the optimal downsampling factor r , we conduct experiments with r set to 1, 2, and 4. As shown in Table VI, the results indicate that kLCRNet performs best with a down-sampling factor r of 2, achieving superior performance compared to other values.

Sampling radius $LOCAL_R$: The sampling radius in the local connectivity exploration module is an important parameter in stage 2. It determines the range within which all potential connection relationships for each keypoint are enumerated to construct the connection relationship set C . Consequently, the choice of $LOCAL_R$ influences the balance between positive and negative samples in C , affecting the overall performance of connectivity classification.

Experimental results for selecting the sampling radius $LOCAL_R$ on the CityScale dataset are presented in Table VII. The APLS index reaches 69.56% with $LOCAL_R = 10$, while the TOPO-F1 score peaks at 78.27% with $LOCAL_R = 12$. These results indicate that $LOCAL_R$ positively correlates with TOPO-P and negatively correlates with TOPO-R. After a comprehensive evaluation, $LOCAL_R = 12$ is chosen as the optimal radius.

Sampling points K : The number of sampling points K plays an important role in the local connectivity exploration module, which establishes connection relationships among keypoints V . As a result, K directly impacts the integrity of the road network. Table VIII presents the experimental results for different choices

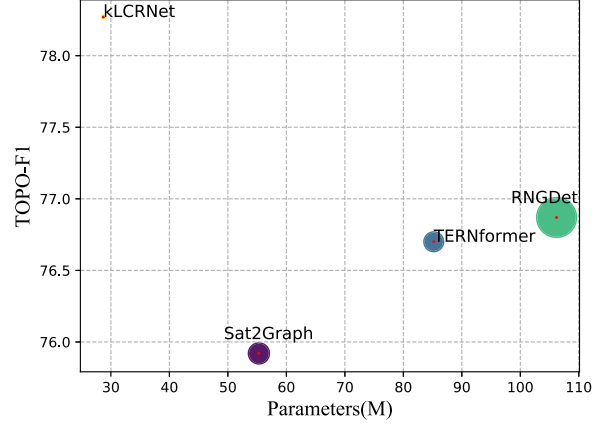


Fig. 8. Model efficiency analysis among different methods. The inference time is denoted by the area of the circle.

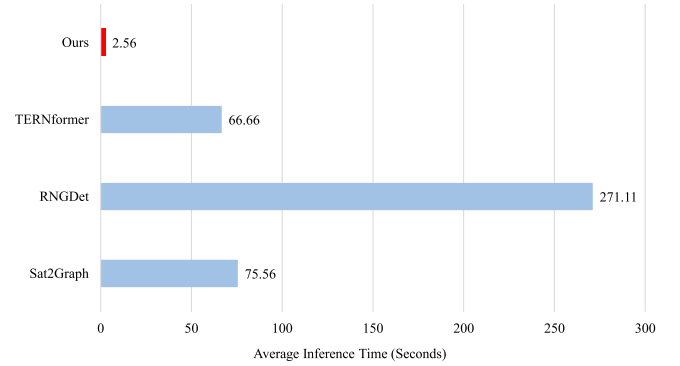


Fig. 9. Inference time on the CityScale Dataset. For a test image with a size of 2048×2048 , kLCRNet is 102 times faster than the iterative detection method RNGDet [21] and 25 times faster than the segmentation method TERNformer [48].

TABLE VIII

COMPARISON OF SAMPLING POINTS K ON THE CITYSCALE DATASET

K	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
2	72.72	52.84	60.49	67.33
4	82.22	50.15	61.57	66.72
6	82.94	49.71	61.43	66.11
8	86.77	71.59	78.27	68.96
10	83.00	49.81	61.53	66.87

TABLE IX

RESULTS FOR WEIGHTING COEFFICIENTS $\lambda_1, \lambda_2, \lambda_3$, AND λ_4 ON THE CITYSCALE DATASET

λ_1	λ_2	λ_3	λ_4	TOPO-P $\uparrow$	TOPO-R $\uparrow$	TOPO-F1 $\uparrow$	APLS $\uparrow$
1.0	1.0	1.0	0.01	84.62	42.04	55.20	50.21
1.0	1.0	1.0	0.1	82.74	49.08	60.89	63.04
1.0	1.0	1.0	1.0	86.77	71.59	78.27	68.96
0.01	0.01	0.01	1.0	84.73	25.75	38.02	29.05
0.1	0.1	0.1	1.0	82.95	50.11	61.66	62.76

of K on the CityScale dataset. The results indicate that when $K = 8$, the model achieves state-of-the-art performance across all evaluation metrics. Therefore, $K = 8$ is selected as the optimal number of sampling points.

Training strategy: kLCRNet is trained in a multitask manner, incorporating keypoint detection, road segmentation, and

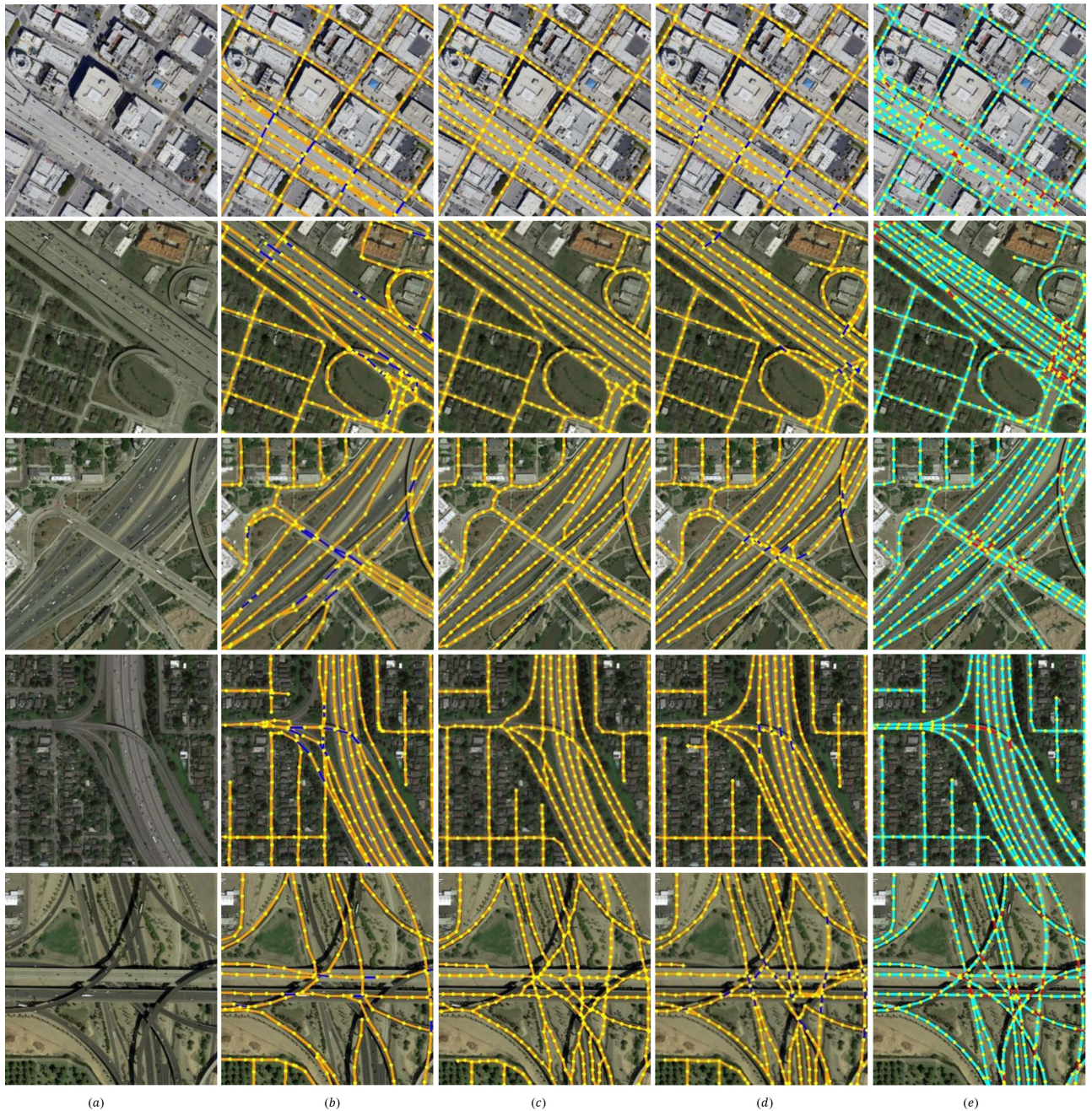


Fig. 10. Visualization of challenging cases on the CityScale dataset. The cyan line represents the ground truth road networks, while the red line indicates the overlap of overpasses in the ground truth. The orange line shows the predicted road networks, and the blue line highlights the predicted overlap of overpasses. The yellow points denote the predicted keypoints.

connectivity classification. The keypoint detection loss consists of keypoint classification loss and offset regression loss, weighted by λ_1 and λ_2 , respectively. The road segmentation loss guides the model to extract rich semantic information for Stage 2. In this ablation study, we determine the optimal weighting coefficients λ_1 , λ_2 , λ_3 , and λ_4 to balance the contributions of different modules. As shown in Table IX, the training objective achieves optimal performance when all weighting coefficients are set to 1.0. Experimental results suggest that each module contributes equally to the final model, which is consistent with findings from other ablation studies.

V. DISCUSSION

In this section, we provide an in-depth analysis of our method, focusing on its model structure and performance. We also discuss its limitations and outline potential directions for future work in the field of road network extraction from remote sensing images.

Efficiency analysis: We evaluate model efficiency in terms of model parameters, accuracy, and inference time. Table X presents the efficiency analysis results on the CityScale dataset at a resolution of 2048×2048 for all comparison methods. As

TABLE X
MODEL EFFICIENCY ANALYSIS

Method	kLCRNet	TERNformer [48]	RNGDet [21]	Sat2Graph [22]
#Params (M) ↓	28.76	85.24	106.19	55.29
Infer. (s) ↓	2.56	66.66	271.11	75.56

#Params (M) and Infer. (s) denote model parameters and inference time, respectively.

shown in Table X, our approach exhibits lower model complexity and higher inference speed compared to existing methods, as kLCRNet is a fully end-to-end and noniterative framework for road network extraction. Fig. 8 provides a comprehensive comparison of model parameters, accuracy (TOPO-F1), and inference time. In this figure, the area of each circle represents the inference time (Infer. (s)). We can see that kLCRNet achieves a notable advantage with lower model complexity and higher accuracy. Specifically, Fig. 9 demonstrates that kLCRNet achieves an average inference time of 2.56 seconds per image, which is 102 times faster than RNGDet [21] and 25 times faster than TERNformer [48]. For images with a resolution of 512×512 , the average inference time is further reduced to just 0.24 s. This significant improvement in computational efficiency enables kLCRNet to meet real-time processing requirements for practical applications.

Difficult cases: Fig. 10 presents several difficult cases, including multiple overlapping overpasses. In the figure, red and blue lines denote overlapping overpasses in the ground truth and predictions, respectively. The visualization highlights our method's significant advantage in handling overpass scenarios. Unlike the semantic segmentation-based method TERNformer [48], kLCRNet leverages a two-stage architecture that first generates road keypoints and subsequently constructs connectivity relationships. By integrating semantic and geometric information, kLCRNet simplifies the road network modeling process while achieving accurate extraction. While our approach outperforms existing methods in extracting road networks in complex scenarios, there remains room for further improvement in capturing intricate scenarios compared to the ground truth.

Limitations and future work: The quantitative results and visualization figures demonstrate that the proposed kLCRNet has achieved excellent results in road network extraction and meets real-time requirements. Nevertheless, some limitations remain: First, kLCRNet struggles with keypoint detection in areas with severe shadows and occlusions, which can lead to incomplete road extraction in complex environments. This indicates that there is still room for improvement in handling such challenging scenes. Second, while kLCRNet can effectively construct connections in cases of overpass overlaps, it faces difficulties with highly complex or completely occluded overpass scenes. The model's performance in these situations is limited by its reliance on top-down remote sensing images, which does not provide depth information or differentiate between overlapping roads at different heights.

VI. CONCLUSION

This article introduces kLCRNet, an innovative efficient road network extraction method that leverages local connectivity

among keypoints. kLCRNet effectively addresses common challenges like connection relationship loss and information degradation. kLCRNet employs a two-stage architecture: the first stage detects and matches road keypoints, while the second stage constructs connectivity relationships between these keypoints. This dual-stage approach combines the high recall rate of semantic segmentation methods and the superior connectivity modeling of graph-based methods. To enhance efficiency, kLCRNet incorporates a local connectivity exploration module that simplifies complex relationship construction into a binary classification problem. In addition, kLCRNet employs a line pooling operation to facilitate efficient, end-to-end construction of road network connectivity. Inference tests on the CityScale dataset reveal kLCRNet's substantial speed advantages, demonstrating its suitability for real-time applications. Further experiments on both the CityScale and SpaceNet datasets validate its effectiveness, especially in complex urban scene. The extensive results show that kLCRNet significantly improves road network extraction, balancing efficiency, and accuracy through the local connectivity exploration of keypoints.

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